

HealthConnect Week 7 Project Summary

Machine Learning Engineering Track

Programme	AnalystLab Africa Experience Lab
Week	Week 7 Project Summary
Project	HealthConnect Clinic Experience Lab
Track	Machine Learning Engineering
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1. What you planned to test

Reproducibility of the Week 6 training pipeline; preprocessing edge cases (first-time patients, missing reminder_channel, unseen categorical values); the model interface's handling of invalid/malformed input; and, once Alia's methodology arrived, whether her one genuinely different feature (booking_month) changes model performance and whether the interface can host a model built on her exact feature set.

2. What you actually tested

12 executed tests across four areas: training reproducibility, preprocessing components, interface input/output validation (including a new malformed-numeric-value test), and two cross-track compatibility tests against a model requiring Alia's booking_month feature. A full retrain with booking_month added was also run and compared against the Week 6 baseline on the identical held-out set.

3. The most important testing results

The Week 6 baseline is exactly reproducible (accuracy 0.629, ROC-AUC 0.683, identical on retrain). A real interface bug was found: a text value in a numeric column crashed with a raw scikit-learn error instead of a clear one. The Week 6 interface could not host a model needing one extra feature at all. Alia's booking_month feature, once tested, did not improve performance (ROC-AUC 0.682 vs. 0.683; accuracy 0.622 vs. 0.629).

4. Issues or weaknesses identified

(1) NoShowModelInterface had no dtype validation on numeric columns, so malformed input surfaced a confusing internal error instead of a clear one. (2) NoShowModelInterface hardcoded a single global feature set, so it could not host a Data Science model whose feature set differs even slightly.

5. Improvements/refinements made

Built NoShowModelInterfaceV2: accepts a per-model feature_fn/feature_cols/numeric_cols contract instead of hardcoded globals, validates numeric dtypes before scoring, and logs scoring activity.

6. Retesting results

Both issues retested and resolved. The malformed-age input now raises a clear, named error identifying the exact column and value. The booking_month model, wrapped in NoShowModelInterfaceV2 with its own feature_fn, scores correctly (3/3 rows, valid probabilities) where it previously failed outright under the Week

6 interface. Of 12 logged tests, 11 passed outright and 1 (the deliberately surfaced Week 6 bug) failed as expected, then passed on retest.

7. Which track(s) you collaborated with

Data Science (Alia Al-Qadri).

8. What was tested collaboratively

Alia's full methodology, confirmed in #hc-pod-03 on 8 and 12 September -- target definition, missing-value handling, feature engineering (including booking_month), excluded predictors, split strategy (GroupShuffleSplit on patient_id, ~80/20, random_state=42), and preprocessing choices. It matches this pipeline almost exactly except for booking_month, which was isolated and tested directly rather than assumed to help.

9. What changed as a result

The model-integration interface was generalised so it can host any model's own feature contract, not just the ML Engineering track's default one -- a real compatibility gap discovered specifically through this comparison. booking_month itself was tested and is not adopted: it does not improve ROC-AUC and slightly lowers accuracy.

10. Key findings or validation outcomes

Pipeline reproducibility confirmed exactly. Two genuine issues found and fixed. Alia's methodology independently validated as matching this pipeline almost exactly, reinforcing Week 6's convergence finding. booking_month tested honestly and found not to help -- consistent with Week 6's tree-model result, reinforcing that the current feature set already captures the available linear signal.

11. Major challenges encountered

Alia's actual fitted model/artifact still hasn't been shared -- she offered a processed dataset and notebook instead, so her methodology was tested directly rather than waiting further. Also hit a recurring Colab _IncompleteInputError from a leftover, never-fixed broken cell in the Week 6 notebook, which silently blocked "Run all" from ever reaching the Week 7 cells until it was found and removed.

12. Important decisions made

Chose not to adopt booking_month, since it produced no measurable improvement rather than assuming a Data-Science-sourced feature must help. Chose to generalise the interface (feature_fn/feature_cols per model) rather than write a one-off fix just for booking_month, since the same gap would recur with any future model.

13. Remaining limitations

No real fitted Data Science model artifact is integrated yet -- only a reproduction of Alia's described methodology. No fairness evaluation has been run on gender/age. The Sunday-appointment / Knowledge Base inconsistency from Week 5 remains open.

14. Remaining issues or dependencies

Awaiting Alia's fitted model or serialized pipeline for a true handoff. The Data Analytics track's segment-level findings have not yet been incorporated into the feature review.

15. Your contribution to the overall HealthConnect project

This week turns the model-integration interface from something that only worked for one hardcoded feature set into infrastructure that can actually host a different track's model on its own terms -- validated concretely against Alia's real, confirmed methodology, not a hypothetical scenario.

16. What must be completed before Week 8

Obtain Alia's actual fitted model/artifact if possible; run full end-to-end validation of the integrated pipeline; incorporate the Data Analytics track's segment findings if shared; complete a fairness pass on gender/age; update README.md to reflect NoShowModelInterfaceV2 as the current integration contract.